PO-0256 Episcleral Brachytherapy for Uveal Melanoma - Reference Center experience

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Purpose or Objective
Uveal melanoma is the most common intraocular tumor in the adult age, and the second most common after skin melanoma, with an annual incidence rate of 4-10 per million people. Portuguese patients with uveal melanoma were sent abroad for treatment until November 2013.
Different therapeutic approaches include enucleation, proton therapy, episcleral brachytherapy (EBT), with similar outcomes.
The purpose of this work was to evaluate treatment response in patients with uveal melanoma submitted to EBT as well as overall survival (OS), recurrence free survival (RFS) and metastasis free survival (MFS).

Materials and Methods
Included all patients with uveal melanoma treated in our institution by the Radiation Oncology Department in co-operation with the Ophthalmology Department, between November 2013 and March 2019. Plaques with low energy photon seeds (^{125}I) were used. Dose prescription was 85 Gy to the tumor apex. Response was evaluated by ophthalmic ultrasound with serial lesion measurements and treatment side effects were recorded. OS, RFS and MFS were estimated using Kaplan-Meier method. A type I error of 0.05 was considered for inferential statistics.

Results
113 EBT procedures were done in 113 patients, with a median age of 61 years, female gender (57,5%) and right eye (51,3%) predominance.Initial median diameter and thickness were 11,54 mm (3,10-18,70 mm) and 6,06 mm (2,00-12,67 mm), respectively. No distant metastatic disease was present at the time of diagnosis.Staging was: Stage IIA 43,4%, Stage IIB 40,7%, Stage I 14,2%, Stage IIIA 1,8%.
Median treatment time was 116,0 hours, median prescribed dose of 86,75 Gy. A notched eye plaque (ROPES15n) was used in 21,2% of cases and a COMS 14 plaque in 21,2%.No complications during implant procedure or treatment period were registered.
Median follow-up was 18 months (1-60 months).During this period, a significant reduction in thickness (median difference of 1,88 mm, $p<0,001$) and diameter (median difference of 1,96 mm, $p<0,001$) was recorded. Radiation retinopathy was observed in 15,9% of patients, treated by intra-ocular anti-VEGF in all cases.
At 24 months the RFS was 95,2%, MFS 86,0% and OS 92,4%.

Conclusion
EBT is an effective and well tolerated eye sparing method to treat patients with uveal melanomas. A progressive significant reduction in tumors diameter and thickness was observed with no severe late side effects registered.

PO-0257 Perioperative Radiation with/without High Dose Rate Brachytherapy for High-risk Soft Tissue Sarcoma

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Purpose or Objective
Evaluation of the therapeutic ratio of maximal perioperative radiation therapy (MRT = external beam radiotherapy with interstitial high dose rate brachytherapy boost) versus standard radiation therapy (SRT) in the form of external beam radiation therapy (EBRT) in patients with high risk soft tissue sarcomas (STS).

Materials and Methods
Retrospective review of STS patients treated between 2009 and 2017 with definitive perioperative radiotherapy with or without interstitial brachytherapy boost. Multidisciplinary evaluation was used to select the highest risk patients for MRT. Chi-square analysis was used to compare MRT and SRT groups, Kaplan-Meier estimates were used for survival and control analysis, and logistic regression was used to evaluate factors predictive of brachytherapy use in our cohort.

Results
Fifty eight patients were identified, 23 (40%) treated with MRT and 35 (60%) SRT. The majority (59%) had grade 3 disease, and 91% had close or positive surgical margins. Median tumor size was 10.5 cm for patients treated with MRT and 6.2 cm for patients treated with SRT. Median brachytherapy dose was 13.5 Gy in 3 fractions over 2 days. After median follow up of 39 months, local control was similar (83%) with MRT and SRT, and no significant difference was observed in distant failure (35% MRT vs 29% SRT, $p=0.62$). Grade 3+ acute toxicity was higher with MRT, 11 (48%) vs 2 (6%) ($p<0.01$), respectively, including 16% (n=9) requiring surgical management. There was no significant difference in grade 3+ chronic toxicity between cohorts, 2 (9%) vs 0 (0%) ($p=0.39$), and no Grade 5 toxicities.

Conclusion
Adding interstitial brachytherapy boost to perioperative external beam radiotherapy for patients with STS did not result in excessive chronic toxicities compared with EBRT alone. Local control was similar with MRT and SRT despite selection of patients with larger tumors in the MRT cohort. The overall impact of interstitial brachytherapy warrants prospective evaluation.